A type is a set exactly when it satisfies Streicher Axiom K

The Univalent Foundations Program

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Editorial note. The counted statement is the author thm environment at source lines 247-251 and its proof environment opens at line 253 after a blank line; the pairing is by adjacency and the retained proof range 253-257 is the whole environment.

Editorial note. The retained counter values (chapter 7, section 2, theorem offset 0) make the counted theorem print as Theorem 7.2.1, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:h-set-uip-K = 7.2.1).

Editorial note. No context span and no external reference are needed: the statement and proof use no label from outside the retained range.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the equivalence between being a set and satisfying Axiom K (author label thm:h-set-uip-K, book Theorem 7.2.1).

Allowed standard prerequisites (editorial).

- Set. X is a set when every path type of X is a mere proposition, equivalently when it satisfies uniqueness of identity proofs.. It is the left hand side of the equivalence.
- Streicher Axiom K. X satisfies Axiom K when for every x and every loop p at x we have p equal to the reflexivity path.. It is the right hand side of the equivalence.
- Path induction. To prove a family of statements about an arbitrary path it suffices to prove the case of the reflexivity path.. It is the only tool used in the converse direction.

Theorem 7.2.1. *A type X is a set if and only if it satisfies **Axiom K**: for all $x : X$ and $p : (x =_A x)$ we have $p = \mathsf{refl}_x$.*

Proof. Clearly Axiom K is a special case of UIP. Conversely, if X satisfies Axiom K, let $x, y : X$ and $p, q : (x = y)$; we want to show $p = q$. But induction on q reduces this goal precisely to Axiom K. □